Business Intelligence

Lecture 2

Lecture Agenda

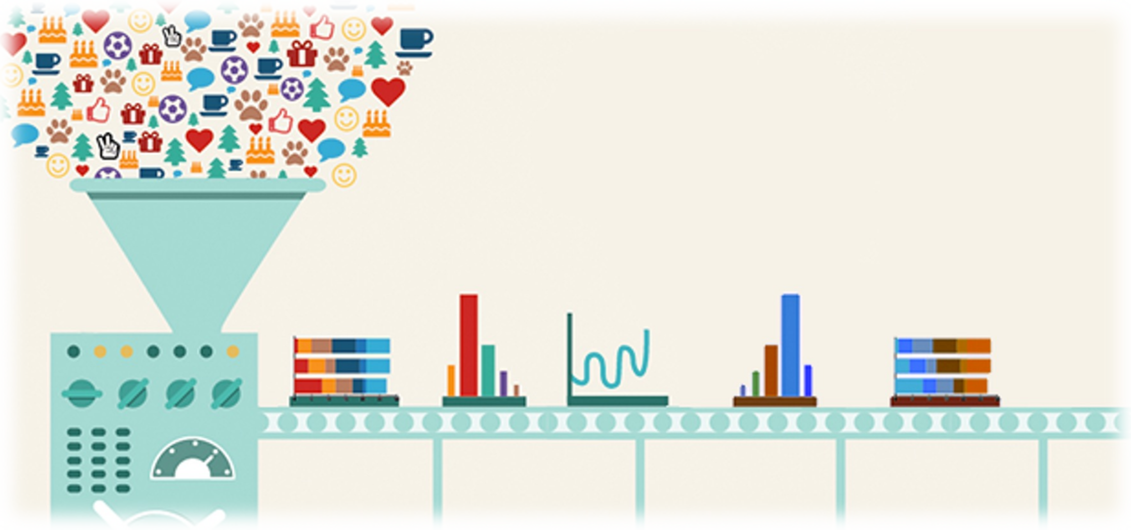
- Data Driven Project (DDP)
- Business Approach to DDP
- Key Performance Indicators (KPI)
- Business Modeling

Data Driven Project

Data Driven Project

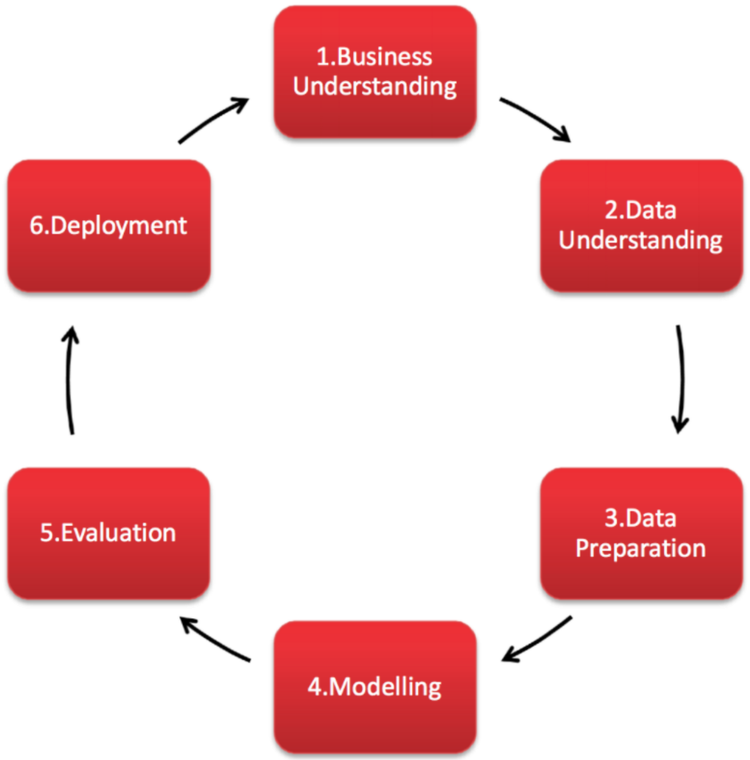
When a company employs a “data-driven” approach, it means it makes **strategic decisions** based on data analysis and interpretation.

A **data-driven approach** enables companies to examine and organize their data with the goal of better serving their customers and consumers.



CRISP-DM

Cross-**I**ndustry **S**tandard **P**rocess for **D**ata **M**ining, known as CRISP-DM, is a model that describes common approaches used by data mining experts. It is the most widely-used analytics model. CRISP breaks the process of data mining into six major phases:



1. Business Understanding

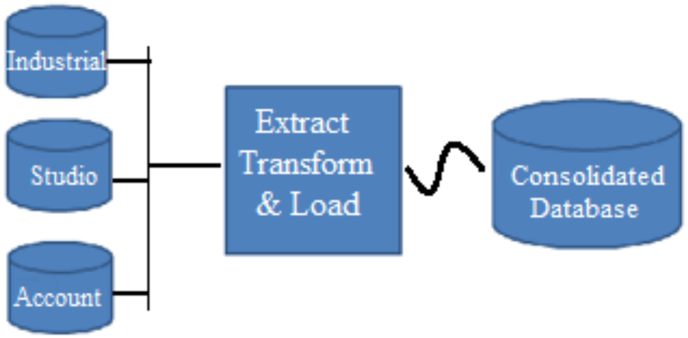
- Understand what you want to accomplish from a business perspective.
- Your organization may have constraints that must be properly balanced.
- Cover important factors that could influence the outcome of the project.
- Bottom line: what is your business goal in this project?

2. Data Understanding

- Acquire the data listed in the project resources.

- This initial collection includes data loading, if this is necessary for data understanding.

If you acquire multiple data sources, then you need to consider how and when you're going to integrate these.



3. Data Preparation

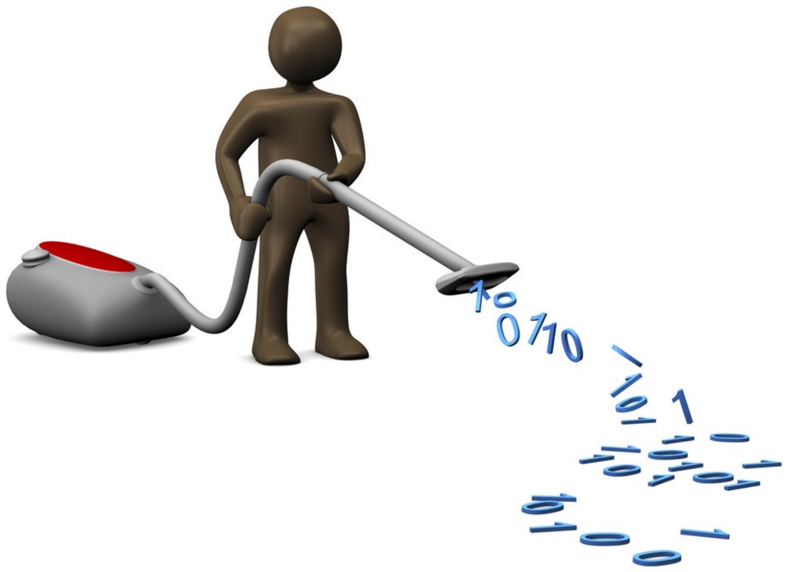
Select This is the stage of the project where you decide on the data that you’re going to use for analysis.

- The criteria you might use to make this decision include the relevance of the data to your data mining goals
- Data selection covers selection of attributes (columns) as well as selection of records (rows) in a table.

3. Data Preparation

<u>Clean</u> This task involves raise the data quality to the level required by the analysis techniques that you’ve selected.

This may involve selecting clean subsets of the data, the insertion of suitable defaults, or more ambitious techniques such as the estimation of missing data by modelling.



3. Data Preparation

Construct This task includes constructive data preparation operations such as the production of derived attributes or entire new records, or transformed values for existing attributes.

Integrate These are methods whereby information is combined from multiple databases, tables or records to create new records or values.

4. Modeling

Technique As the first step in modelling, you'll select the actual modelling technique that you'll be using. Although you may have already selected a tool during the business understanding phase, at this stage you'll be selecting the specific modelling technique e.g. decision-tree, clustering, neural networks, etc.

If multiple techniques are applied, perform this task separately for each technique.

4. Modeling

Build run the modelling tool on the prepared dataset to create one or more models.

Assess your data mining success criteria and your desired test design.

- Judge the success of the application of modelling and discovery techniques technically
- Contact business analysts and domain experts to discuss the data mining results
- This task only considers **models**, whereas the evaluation phase also takes all other results that were produced in the project.

5. Evaluation

Determine if there is some business reason why this model is deficient.

- The evaluation phase involves assessing any other data mining results you've generated.
- Data mining results involve models that are necessarily related to the **original business objectives**
- might also include additional challenges, information, or hints for future directions.

6. Deployment

In the deployment stage you'll take your evaluation results and determine a strategy for their deployment.

- This is where predictive analytics really helps to improve the operational side of your business.
- **Final Report** - depending on the deployment plan, this report may be only a summary of the project and its experiences, or it may be a final and comprehensive presentation of the data mining result(s).

Example

The sinking of the **Titanic** is one of the most infamous shipwrecks in history. On April 15, 1912, during her maiden voyage, the Titanic sank after colliding with an iceberg, killing 1502 out of 2224 passengers and crew **(32% survival rate)**.

Example

One of the reasons that the shipwreck led to such loss of life was that there were not enough lifeboats for the passengers and crew. Although there was some element of luck involved in surviving the sinking, **some groups of people were more likely to survive than others, such as women, children, and the upper-class**.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Business understanding - what characterizes the survivors?

Example

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
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The problem is a **binary class classification** problem, so we need to choose the right model for this problem.

For example, linear regression is not a good choice.

Key Performance Indicator (KPI)

What is A KPI

- A KPI (Key Performance Indicator) is a metric that is tied to a target.
- A KPI represents how far a metric is above or below a pre-determined target.
- Choosing the right KPI is crucial to make effective, data-driven decisions.
  - Right KPI will help to concentrate the efforts of employees towards a meaningful goal
  - Wrong KPI could waste significant resources chasing after vanity metrics

A great KPI is SMART (**S**pecific, **M**easurable, **A**chievable, **R**elevant, **T**ime phased)

The Five Conditions of good KPI's - SMART

Specific

It has to be clear what the KPI **<u>exactly</u>** measures

Measurable

The KPI has to be measurable to define a standard, budget or norm, to make it possible to measure the actual value and to make the actual value comparable to the budgeted value

Achievable

"Nothing is more discouraging than striving for a goal that you will never obtain"

The Five Conditions of good KPI’s - SMART

Relevant

- Must give more insight in the performance of the organization
- If a KPI is not measuring a part of the strategy, acting on it doesn’t affect the organizations’ performance.
- An irrelevant KPI is useless.

Time phased

It is important to express the value of the KPI in time

How To Achieve Success?

- Identify areas of activity that require greater attention
- Performance indicators that are grounded (SMART)
- Agile development
- Business understanding

KPI vs Metrics

- **KPIs are the key targets you should track** to make the most impact on your strategic business outcomes. KPIs support your strategy and help your teams focus on what's important.
- **Metrics measure the success** of everyday business activities that support your KPIs. While they impact your outcomes, they're not the most critical measures.

Why Are KPIs Important?

- **Keep your teams aligned**

  Whether measuring project success or employee performance, KPIs keep teams moving in the same direction.

- **Provide a health check**

  Key performance indicators give you a realistic look at the health of your organization, from risk factors to financial indicators.

Why Are KPIs Important?

- **Make adjustments**

  KPIs help you clearly see your successes and failures so you can do more of what's working, and less of what's not.

- **Hold your teams accountable**

  Make sure everyone provides value with key performance indicators that help employees track their progress and help managers move things along.

Examples of KPIs

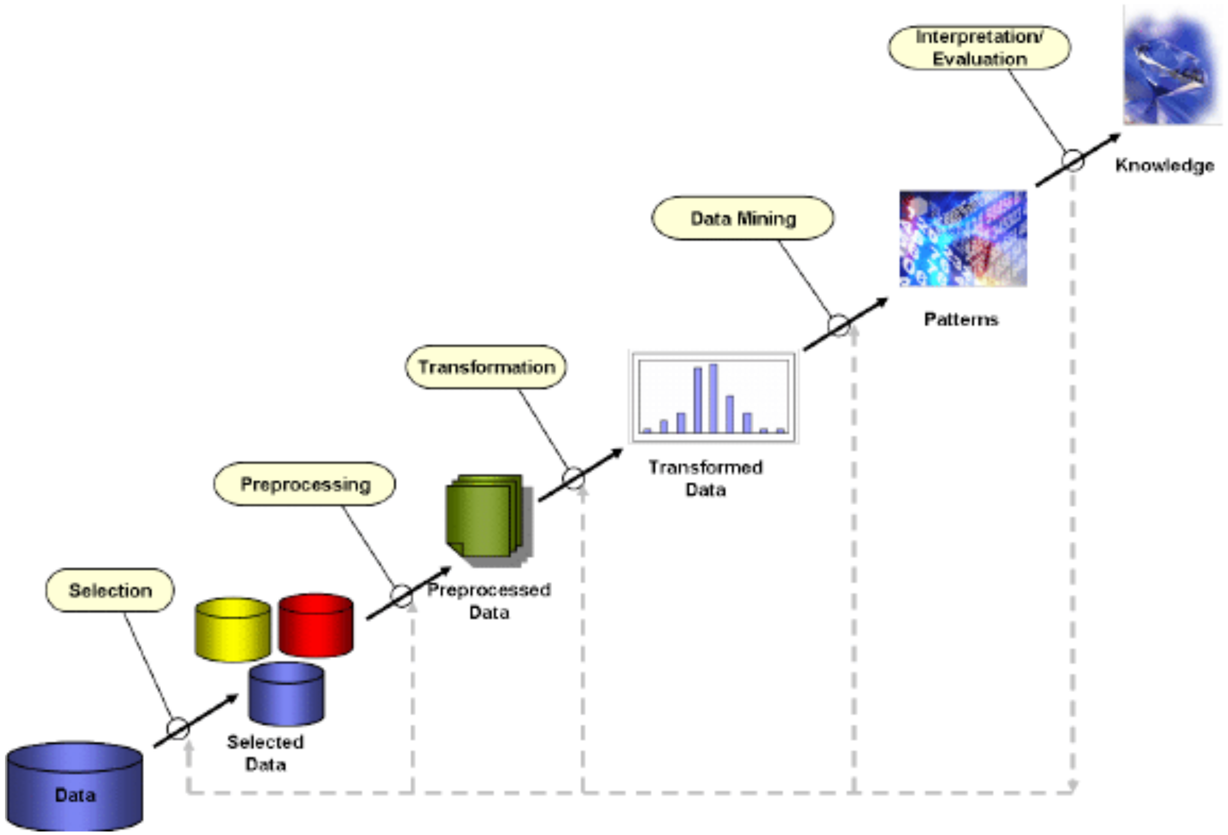
- **Sales**: number of new contracts signed per period
- **Financial**: growth in revenue
- **Customers**: products usage, customers satisfaction
- **Operational**: TTM – Time To Market
- **Marketing**: monthly customers' traffic

Data Mining for Decision Making

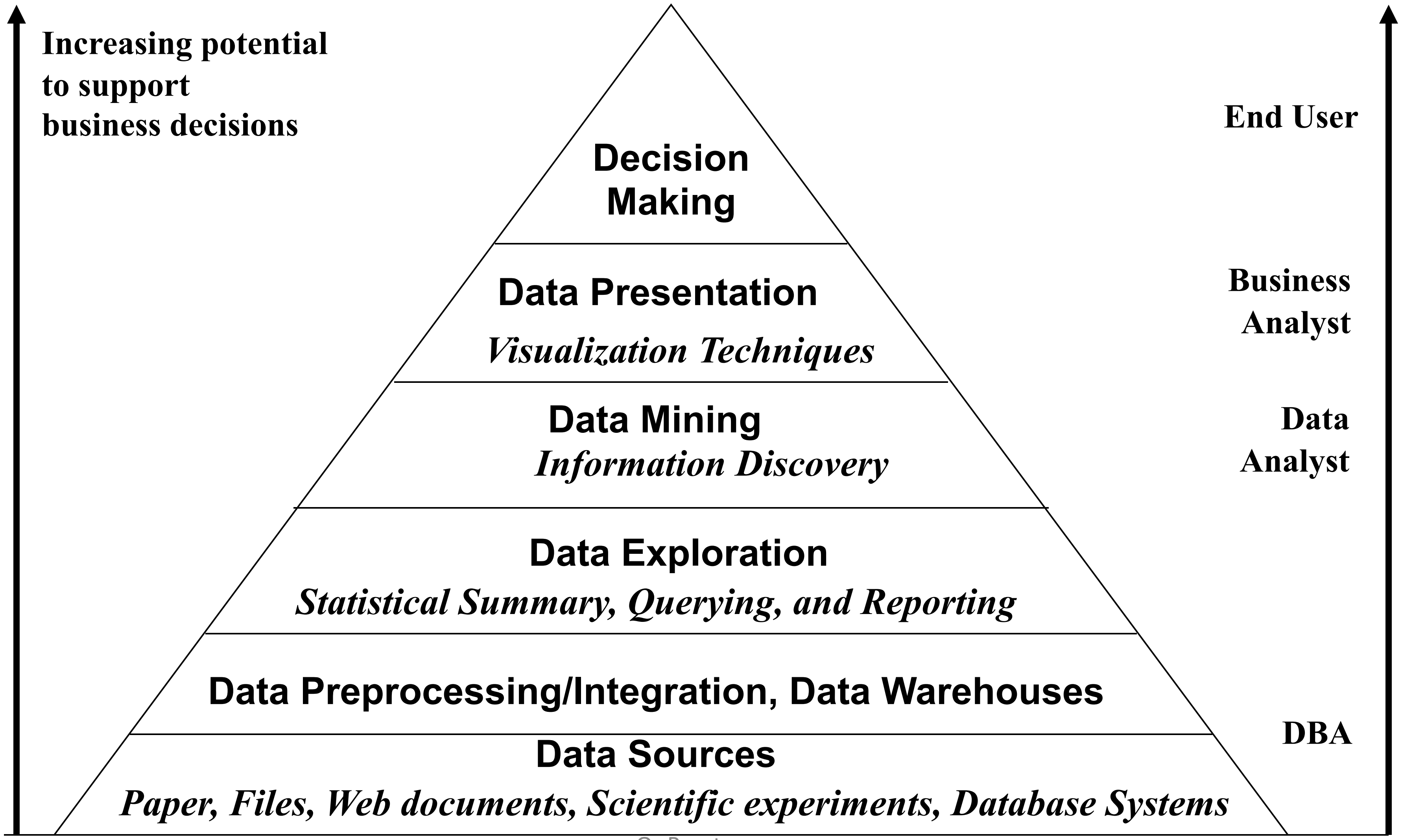
What Is Data Mining?

- Knowledge discovery (mining) in databases (KDD)
- knowledge extraction
- data/pattern analysis
- data archeology
- information harvesting

Knowledge Discovery (KDD) Process



Data Mining and Business Intelligence



Applications

- Market analysis and management

  Target marketing, customer relationship management (CRM)

- Risk analysis and management
  - Forecasting, customer retention, improved underwriting, competitive analysis
- Fraud detection
- Detection of unusual patterns (outliers)

Example - Market Analysis and Management

- Where does the data come from?

  Credit card transactions, loyalty cards, discount coupons, customer complaint calls, surveys …

- Target marketing
  - **Find clusters of “model” customers who share the same characteristics.** For example, most customers with income level 60k – 80k with food expenses $600 - $800 a month live in that area.
  - **Determine customer purchasing patterns over time**. For example, customers who are between 20 and 29 years old, with income of 20k – 29k usually buy this type of CD player

- Cross-market analysis

  customers who buy computer A usually buy software B

Example - Market Analysis and Management

- **Customer requirement analysis**

  Identify the best products for different customer, Predict what factors will attract new customers

- **Summary information**
  - Multidimensional summary reports
    - Summarize all transactions of the first quarter from three different branches
    - Summarize all transactions of last year from a particular branch
  - Statistical summary information
    - What is the average age for customers who buy product A?
- **Fraud detection**: find outliers of unusual transactions
- **Financial planning**: summarize and compare the resources and spending

Types of Data

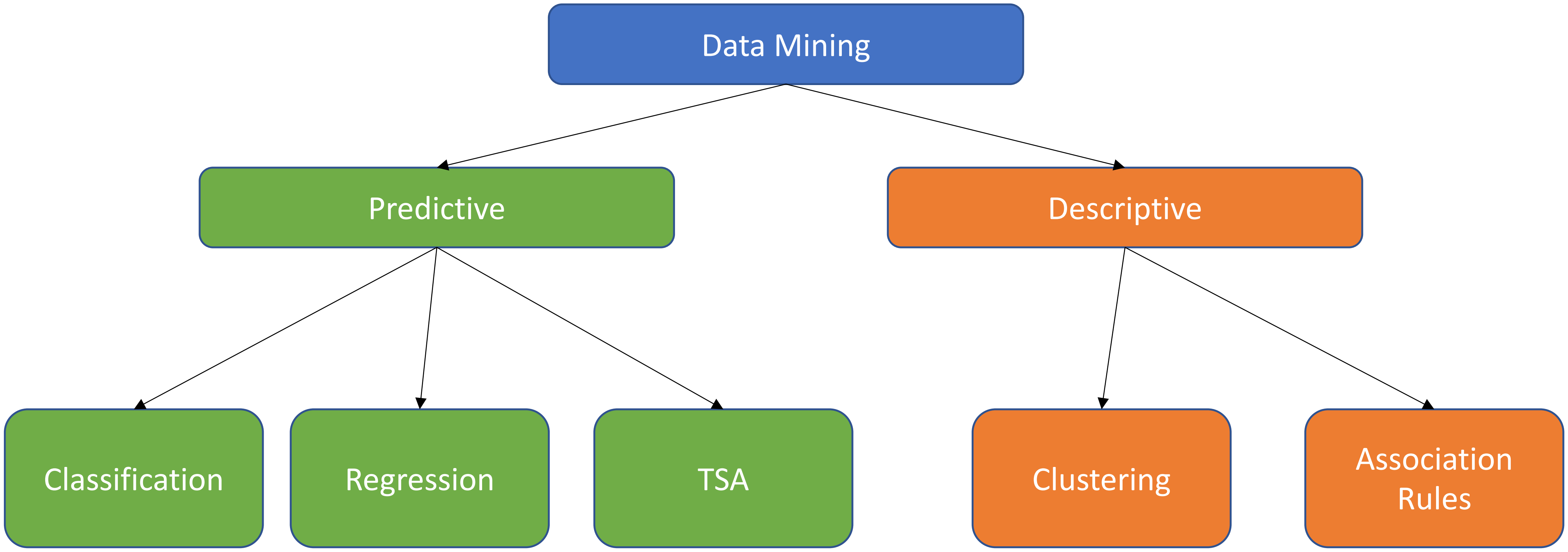
Database-oriented data sets and applications

- Relational database, data warehouse, transactional database

Advanced data sets and advanced applications

- Object-Relational Databases
- Time-Series databases
- Spatial Databases
- Text databases and Multimedia databases
- Data Streams
- The World-Wide Web

Data Mining Models and Tasks



SQL – DML and DDL

SQL - Structured Query Language

- Query Language
- Table Oriented
- Fast and Easy access to our data
- CRUD - Create, Read, Update, Delete

SQL

DML - Data Manipulation Language

a. Select
b. Insert
c. Update
d. Delete

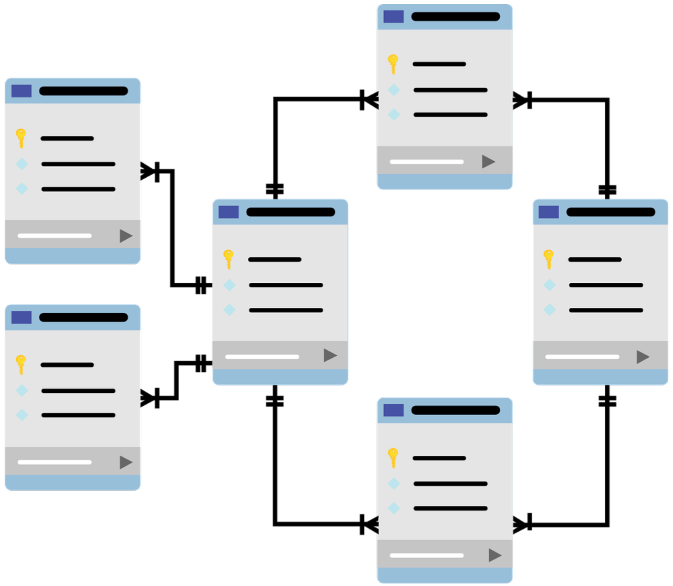
DDL - Data Definition Language

a. Create
b. Drop
c. Alter

Overview

What happens when we want to handle a few tables which relative

one to each other ?



Foreign Key

Courses			
StudentID	CourseNumber	CourseName	Grade
111	289	DB	96
111	281	Algo	85
222	281	Algo	78

Students		
FirstName	LastName	StudentID
Avi	Cohen	111
Dan	Israeli	222
Ofer	Bar	333

Primary Key

How to merge two tables ? - Option 1

SELECT FirstName, CourseName, Grade

FROM Students as S, Courses as C

Where (S.StudentID = C.StudentID) and (C.CourseName = ‘Algo’)

How to merge two tables ? - Option 2

SELECT FirstName, CourseName, Grade

FROM Students as S JOIN Courses as C

ON S.StudentID = C.StudentID

Where (C.CourseName = ‘Algo’)

ON VS. USING

- ➔ When using JOIN statement we need to understand according to which column we want to join the tables.
- ➔ We will use ON when the columns name is not equal.
- ➔ We will use USING when the columns name is equal.

How to merge two tables ? - Option 3

SELECT FirstName, CourseName, Grade

FROM Students JOIN Courses

USING (StudentID)

Where (CourseName = ‘Algo’)

FirstName	CourseName	Grade
Avi	Algo	85
Dan	Algo	78

JOIN Types

- INNER JOIN
- OUTER JOIN
    - LEFT OUTER JOIN
    - RIGHT OUTER JOIN
    - FULL OUTER JOIN

Let's start with an example..

Given 2 Table called R1, R2:

Mutual column is "Name"

R1:

Name	Exam
Avi	85
Or	73
Dan	98

R2:

Name	Exercises
Or	95
Dan	90
Liat	83

LEFT OUTER JOIN

What happens here? Why Avi has “null” value?

What happens if we use only “JOIN”?

Name	Exam	Exercise
Avi	85	Null
Or	73	95
Dan	98	90

RIGHT OUTER JOIN

What happens here? Why Avi has “null” value?

What happens if we use only “JOIN”?

Name	Exam	Exercise
Or	73	95
Dan	98	90
Liat	Null	83

FULL OUTER JOIN

What happens here? Why Avi has “null” value?

What happens if we use only “JOIN”?

Name	Exam	Exercise
Avi	85	Null
Or	73	95
Dan	98	90
Liat	Null	83

Handling NULL values

- Sometimes part of our data is NULL, from many reasons:
  - Result of left/right/full outer join
  - Missing data while insert a new row
  - etc.
- What if we want to know who is the problematic data?

Example

Given 2 simple tables (Student and Courses, the favorite example):

Students		
FirstName	**LastName**	**StudentID**
Avi	Cohen	111
Dan	Israeli	222
Ofer	Bar	333

Courses			
StudentID	**CourseNumber**	**CourseName**	**Grade**
111	**289**	**DB**	**96**
111	**281**	**Algo**	**85**
222	**281**	**Algo**	**78**

Example

SELECT FirstName, LastName, CourseNumber

FROM Student LEFT OUTER JOIN Courses

USING (StudentID)

FirstName	LastName	CourseNumber
Avi	Cohen	289
Avi	Cohen	281
Dan	Israeli	281
Ofer	Bar	Null

Example

We store the previous query on new table called NewCourses.

```
SELECT *

FROM NewCourses

WHERE CourseNumber is NULL
```

FirstName	LastName	CourseNumber
Ofer	Bar	Null

Important Notes

- ➔ As we use “is NULL” , we can use “is NOT NULL”.
- ➔ Wrong Usage: FirstName = ‘NULL’

NULL is not a String!

CREATE Tables

```
CREATE TABLE table_name
(
    column_name1 data_type,
    column_name2 data_type,
    column_name3 data_type,
    ...
)
```

1 - Define your table schema + types

```
CREATE TABLE Students
(
        Id int NOT NULL,
        First_Name text,
        Last_Name text,
        Birth_Date date,
        Hour_Salary float
)
```

2 - Insert Values (with the same order!)

INSERT INTO Students

VALUES(123456789, ‘Israel’, ‘Israeli’, ‘1980-03-20’, 120.40)

3 - Check your Insertion

SELECT *

FROM Students

Students				
Id	First_Name	Last_name	Birth_date	Hour_Salary
0123456789	Israel	Israeli	20/03/1980	120.40

SubQuery with ALL

We want to find the city name where all amounts bigger (or equals) than all average amount of all cities.

```
SELECT City
FROM Students
GROUP BY City
HAVING AVG(Amount) >= ALL (SELECT AVG(Amount)
                           FROM Students
                           GROUP BY City)
```

2400
3000
2100
1000

City
Tel-Aviv

SubQuery with IN

We want to find all names which lives in one of the following cities: Eilat, Haifa, Jerusalem.

```
SELECT Name

FROM Students

WHERE City IN (‘Eilat’, ‘Haifa’, ‘Jerusalem’)
```

Name
Haim Itzhak

SubQuery with EXIST

What is the output of the following query?

```
SELECT SUM(Amount)
FROM Students
WHERE EXISTS (SELECT *
              FROM AgudaMembers as AM
              WHERE AM.Name = Students.Name)
```

SUM(Amount)
13,100